

Global Warming Science

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Greenhouse

Please use the template below to answer the workshop questions. “XX” indicates places where you need to complete/write code or add a discussion.

your name: first last

```
[ ]: import numpy as np
      from numpy import linalg
      import matplotlib.pyplot as plt
      import matplotlib
      import pickle
      from matplotlib.ticker import MultipleLocator
      import scipy as scipy
      from scipy import optimize
      import warnings

      # load the data from a pickle file:
      with open('./greenhouse_variables.pickle', 'rb') as file:
          d = pickle.load(file)
          # print information about each extracted variable:
          for key in d:
              print(
                  f"extracting pickled variable: "
                  f"name={key}; size={d[key].shape}"
              )
              # print(f"; type={type(d[key])}")
          globals().update(d)
```

Explanation of variables:

CO2__observed is a time series of observed CO₂, as a function of time in years given by variable **CO2__observed__years**.

Similarly, **CO2__rcp*** are time series of projected CO₂ according to different RCP scenarios.

OLR_280ppm_CO2__only is the top-of-the-atmosphere outgoing longwave radiation, with 280 ppm of CO₂, as function of **wavenumbers**, that are given by the variable of that name. similarly for **OLR_560ppm_CO2__only**.

1) Observed and projected increase in greenhouse gasses:

(a) Plot CO₂ since 1850, superimpose the predicted CO₂ concentration from the RCP8.5 (sometimes referred to as “business as usual”) scenario from a previous IPCC report going to 2100.

```
[ ]: fig = plt.figure(1,figsize=(3,3),dpi=200)
      #plt.plot(XX,XX,lw=3,color="blue",label="observed CO2")
      #plt.plot(XX,XX,lw=1,color="red",label="Projected rcp8.5 CO2")
      plt.legend()
      #plt.xlabel(XX)
      plt.ylabel('CO$_2$ Concentration (ppm)');
```

1(b) An extra challenge problem; if you choose to do this one, skip 1(a). Download and plot the observed CO₂ emissions and CO₂ concentration from 1850 to today. Plot each of the two, together with those from all SSP scenarios to 2100. Evaluate which scenario is closest to the observed, and attempt to explain the observed signal since the 2010s.

```
[ ]: XX
```

2) Energy balance model:

- (a) Calculate the downwelling LW and absorbed SW at the surface in the two-layer model, using an albedo of 0.25, an atmospheric emissivity of 0.8, and searching the web for the other constants.

```
[ ]: # constants:
SO = XX # solar constant
albedo = XX # Albedo
epsilon = XX # emissivity
sigma = XX # stefan-boltzmann constant

# calculate atmospheric temperature T_a:
T = XX
T_a=XX
print(f"T = {T:g}, T_a = {T_a:g}")
# answer:
LW_down=XX
SW_net=XX
print(f"LW_down = {LW_down:g}, SW_net = {SW_net:g}")
```

- (b) Modify the 2-layer energy balance model to take into account the absorption of a fraction $\Delta = 0.15$ of the downward shortwave radiation by the atmosphere. Calculate the resulting temperatures, compare to the case of $\Delta = 0$. Use the results to discuss how increased atmospheric aerosol concentrations due to industrial emissions, which increases SW absorption, can mask the surface warming due to CO₂ increase.

Solution

The new energy balance equation for the surface is:

$$\frac{S_0}{4} XX(1 - \alpha) + \epsilon \sigma T_a^4 = \sigma T^4 \quad (1)$$

The new energy balance equation for the atmosphere is:

$$XX + \epsilon \sigma T^4 = 2\epsilon \sigma T_a^4 \quad (2)$$

where T is the surface temperature, and T_a is the atmospheric temperature. Substituting the second equation into the first yields:

$$XX = \sigma T^4 \quad (3)$$

Solving for T yields:

$$T = \left[\frac{XX}{\sigma(1 - \frac{\epsilon}{2})} \right]^{1/4} \quad (4)$$

```
[ ]: # evaluating the solution numerically:
```

```
Delta = .15
```

```
T = (XX)**(1/4)
print(f"T = {T:g}, in Celsius: {T-273.15:g}")
T_a = XX
print(f"T_a = {T_a:g}")
```

Discussion:

XX

- (c) Consider an abrupt CO₂ doubling corresponding to the transition from panel a of Figure 2.4 to panel b. Considering the response of an atmosphere with a continuous vertical temperature profile of a prescribed lapse rate, calculate the increase in the emission height corresponding to an increase in radiative forcing by 4 W,m⁻² assuming a lapse rate of 6.5 K/km.

```
[ ]: # first, the temperature decrease required to lower the LW by 4 W/m^2:

# The emission height temperature is found by the balance of outgoing
# long-wave radiation from the emission height balancing the incoming solar
# radiation minus the refelected part.
T=XX
# Calclate the right value of DeltaT that leads to a 4 W/m^2 change in F:
DeltaT=XX
print(f"DeltaT = {DeltaT:g}")
# check that radiative forcing changed as expected:
print(f"DeltaF = {sigma * (XX)**4 - sigma * XX**4:g}")

# now, given a lapse rate of 6.5 K/km, this implies the following
# change in emission height:
print(f"Change in emission height: {XX:g} m")
```

3) Radiative forcing

- Explain the upper and lower boundaries of the trapezoidal gap in the shape of the OLR curve around 600-800 cm⁻¹ (Figure 2.5b) due to CO₂ absorption, in terms of the emission height.
- Plot the OLR in a standard atmosphere in the presence of 280 and 560 ppm CO₂, ignoring all other greenhouse gasses.
- Estimate the radiative forcing of 280 ppm of CO₂ by fitting a trapezoid to your plotted OLR curve and calculating its area.
- Calculate the radiative forcing due to a doubling of the CO₂ from the preindustrial value of 280 ppm, by integrating over the difference of the two OLR curves.

Solutions

a) XX

```
[ ]: # (b)

# Read in pickle data of the two OLR curves and plot
fig = plt.figure(1,figsize=(6,4),dpi=300)
ax = plt.gca()
plt.plot(XX,XX,lw=1,color="blue",label="280 ppm")
plt.plot(XX,XX,lw=1,color="red",ls='--',label="560 ppm")
plt.xlabel('Wavenumber (cm$^{-1}$)')
```

```
plt.ylabel('OLR (W/[m$^2$cm$^{-1}$])')

# plot tickmarks and grid
ax.xaxis.set_minor_locator(MultipleLocator(50))
ax.yaxis.set_minor_locator(MultipleLocator(.02))

# title and legend:
plt.title("OLR as function of wavenumber")
plt.grid(which='both',lw=0.25)
plt.legend();
plt.show();
```

- c) We choose the top of the trapezoid as XX. This gives a total height of the trapezoid as XX W/m². The bottom width is about XX cm⁻¹ and the top width is about XX cm⁻¹. Thus using the formula for a trapezoid, $A = H \times (Base_1 + Base_2)/2$ we get a radiative forcing of XX W/m².
- d) The integral over the OLR curve tells you the total outgoing longwave radiation. The difference between the total OLR with 280ppm and 560ppm gives you the radiative forcing of doubling CO₂.

```
[ ]: # approximate the integral by a sum:
Delta_wavenumber = wavenumbers[1] - wavenumbers[0]
RF = Delta_wavenumber*np.sum(XX-XX)
print(f"Radiative Forcing: {RF:.3f} W/m^2")
```

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[ ]:
```

4) Logarithmic dependence

If a doubling of CO₂ from 280 ppm to 560 leads to, say, 3 °C warming, what warming do you expect at an equivalent CO₂ mixing ratio of 500 ppm? Explain why the warming by 2025 is only about 1.2 °C.

```
[ ]: # if the warming for CO2 doubling is 3 degree, we can write the warming as
C02=560
DeltaT=XX
print(f"C02 = {C02:g}, DeltaT = {DeltaT:g}")
C02=500
DeltaT=XX
print(f"C02 = {C02:g}, DeltaT = {DeltaT:g}")
```

Discussion: XX

5) Global warming potential:

Suppose the GWP of methane for a time horizon of 100 yr is 34. Assume methane concentration decays exponentially over 12.4 yr (so the decay function is $e^{-t/12.4}$) while that of CO₂ decays over 200 yr (e.g., due to dissolution in the ocean; note that while such a timescale is often used, the actual expected residence time of anthropogenic CO₂ in the climate system is thousands of years.) (a) How much more efficient is a kilogram of methane at absorbing LW radiation than a kilogram of CO₂? (b) Explain the difference between GWP and absorption efficiency. (c) What is the methane GWP for a time horizon of 500 yr? Explain.

- (a) Reformulate the GWP equation to solve for a_x/a_r .

XX

Therefore, we can conclude that kg methane absorbs more than XX times more strongly than a kg of CO₂.

- (b) XX discuss

(c) XX

6) Water vapor feedback to increased CO₂:

Modify the energy balance model equations (2.2) to include the effects of increasing water vapor and calculate the warming as a function of CO₂ with and without this feedback as follows. Assume the atmospheric emissivity depends on CO₂ and moisture $q(T_a)$ as $\epsilon(\text{CO}_2, q(T_a)) = \epsilon_0 + A \log_2(\text{CO}_2/280) + B \log_2(q(T_a)/q(T_{a0}))$, where $\epsilon_0 = 0.75$ and T_{a0} is the atmospheric temperature for CO₂=280 ppm, and that the atmospheric relative humidity is the same before and after the CO₂ change. Find values for A and B such that the warming due to CO₂ doubling only is 2°C, and due to both feedbacks is 4°C. Plot $T_a(\text{CO}_2)$ for CO₂ in the range of 200 to 800 ppm with and without the water vapor feedback.

```
[ ]: # constants:
albedo = 0.32
epsilon_0 = 0.75
Pa=500 # mid-atmospheric pressure

# atmospheric temperature at 280 ppm:
T_a0 = XX

# first, find what changes to the emissivity are needed to cause the suggested
# warmings:
# Rerun this cell with different values of A until the temperature rise due to CO2
# doubling alone is 2 K:
A = XX # change due to CO2 doubling
T_a_2XCO2 = XX # energy balance solution in terms of an emissivity of epsilon_0+A
print(f"T_a0 = {T_a0:g}, T_a_2XCO2 = {T_a_2XCO2:g}")
```

Now find B:

```
[ ]: # help functions: emissivity and saturation specific humidity:
def emissivity(CO2,T_a,A,B):
    # emissivity a function of CO2 and atm temperature
    epsilon=XX
    return epsilon

def q_sat(T,P):
    # saturation specific humidity (g water vapor per g moist air):
    # inputs:
    # T: temperature, in Kelvin
    # P: pressure, in mb
    R_v = XX # Gas constant for moist air = 461 J/(kg*K)
    R_d = XX # Gas constant 287 J K^-1 kg^-1
    TT = XX # Kelvin to Celsius
    qw=
    return qw

# the desired temperature with both feedbacks as specified in the question:
T_a_2XCO2_water_vapor = T_a_2XCO2+2;

# now calculate and plot the temperature as a function of the coefficient B to find
# when it reaches the assumed warming:
```

```

Brange = np.arange(XX)
Trange=XX # a array with zeros of the same dimension of Brange, for
          # tempeprature as a function of B
i=-1
for B in Brange:
    i=i+1
    epsilon = XX
    Trange[i]=XX # energy balance solution with appropriate emissivity

# Plot:
fig = plt.figure(figsize=(6,3),dpi=300)
plt.plot(XX \
         , label='temperature with water vapor feedback')
plt.plot(XX \
         , label='T_a_2XCO2_water_vapor')
plt.legend()
plt.xlabel('XX')
plt.ylabel('XX')
plt.grid(lw=0.25)
plt.title('XX');
plt.show();

```

Use A,B to solve for Ta and plot with and without the water vapor feedback:

```

[ ]: # from the plot, we conclude that
B=XX
# results in the desired temperature response to CO2 and water vapor.

# Now plot the temperature with and without the water vapor
# feedback as a function of CO2.

# helper function to solve for T_a when emissivity depends on it:
def solve_for_T_a(T_a):
    # calculate
    #  $T_a - ((S_0/4) * (1 - albedo) / (\sigma * (2 - \epsilon(CO_2, T_a))))^{1/4}$ , to
    # be called by scipy.optimize.fsolve in order to find a solution
    # for T_a that satifies this equation
    equation=XX
    return equation

# range of CO2 to consider:
CO2_range=np.arange(XX)
# Prepare arrays for atmospheric temperature response to CO2
# and to both CO2 and water vapor:
Ta_CO2_only=XX
Ta_both=XX

i=-1
for CO2 in CO2_range:
    i=i+1
    B=0; epsilon=emissivity(XX,XX,XX,XX)
    Ta_CO2_only[i]=XX
    B=XX # from previous cell
    x0=XX # initial guess

```

```
T_a=optimize.fsolve(solve_for_T_a,x0,args=(A,B))
Ta_both[i]=T_a.item()
```

```
# Plot Ta vs CO2 with and without water vapor feedback:
fig = plt.figure(figsize=(6,3),dpi=300)
plt.plot(CO2_range, XX, color = 'orange', label='CO$_2$ only')
plt.plot(CO2_range, XX, color = 'red', label='CO$_2$ and water vapor')
plt.legend()
plt.xlabel('XX')
plt.ylabel('XX')
plt.grid(lw=0.25)
plt.title('XX');
plt.show();
```